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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 6.3

Orthogonal Projections

Orthogonal Projections – Last Time

Goal: Decompose a vector $\mathbf{y}$ in $\mathbb{R}^n$ into the sum of two vectors, one of which is a multiple of some nonzero vector $\mathbf{u}$ in $\mathbb{R}^n$ and the other orthogonal to $\mathbf{u}$.

That is, we wish to write

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α and $\mathbf{z}$ is orthogonal to $\mathbf{u}$.

Orthogonal Projections – Last Time

If $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ where $\hat{\mathbf{y}} = \alpha \mathbf{u}$, then

- $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$
- $\mathbf{z}$ is the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$

If L is the subspace spanned by $\mathbf{u}$, $\hat{\mathbf{y}}$ is also called the orthogonal projection of $\mathbf{y}$ onto L .

Specifically,

$$\hat{\mathbf{y}} = \text{proj}_L \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

Orthogonal Projections – Today

Goal: Decompose a vector $\mathbf{y}$ in $\mathbb{R}^n$ into the sum of two vectors, one of which is in a subspace W of $\mathbb{R}^n$ and the other of which is in $W^\perp$.

That is, we wish to write

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

Theorem – Last Time

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. Then, each $\mathbf{y}$ in W can be written as

$$\mathbf{y} = c_1 \mathbf{u}_1 + \dots + c_p \mathbf{u}_p$$

where the weights are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then, each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is any orthogonal basis for W , then

$$\hat{\mathbf{y}} = \text{proj}_W \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \dots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

Example 1

Given that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set, find the orthogonal projection of $\mathbf{y}$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

$$\mathbf{y} = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Example 2

Given that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set, let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ and write $\mathbf{y}$ as the sum of a vector in W and a vector orthogonal to W .

$$\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$$

The Best Approximation Theorem

Let W be a subspace of $\mathbb{R}^n$, let $\mathbf{y}$ be any vector in $\mathbb{R}^n$ (not necessarily in W), and let $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W . Then, $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$.

Example 3

Find the distance from $\mathbf{y}$ to the plane in $\mathbb{R}^3$ spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$.

$$\mathbf{y} = \begin{bmatrix} 5 \\ -9 \\ 5 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} -3 \\ -5 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

Example 4

Let $\mathbf{y} = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$, $\mathbf{u}_1 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$

and let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

If $U = [\mathbf{u}_1 \ \mathbf{u}_2]$, compute $U^T U$ and $U U^T$.

Then, compute $\text{proj}_W \mathbf{y}$ and $(U U^T) \mathbf{y}$.

Theorem

If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbb{R}^n$, then

$$\text{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1) \mathbf{u}_1 + \dots + (\mathbf{y} \cdot \mathbf{u}_p) \mathbf{u}_p$$

If $U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_p]$, then
 $\text{proj}_W \mathbf{y} = (U U^T) \mathbf{y}$

for all $\mathbf{y}$ in $\mathbb{R}^n$.
